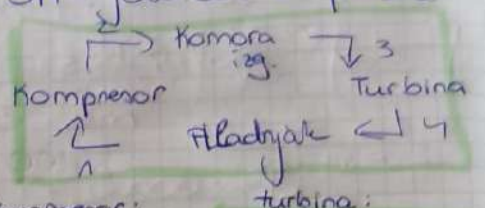
(12.) desnokretni Brayton-Jouleov kr. proces

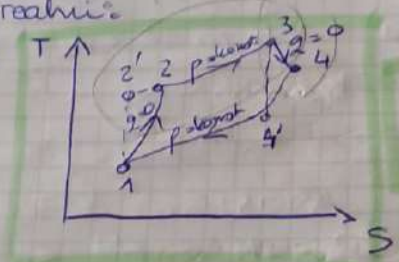
$K = 1,4$

$R = 287\ J/kgK$



→ realni:

a)



Kompresor:

$$\eta_{ik} = \frac{W_{id}}{W_{red}}$$

turbina:

$$\eta_{it} = \frac{W_{red}}{W_{id}}$$

$\vartheta_1 = 15°C = 288,15K$

$p_1 = p_4 = 1\,bar = p_4'$

$p_2 = p_3 = 5\,bar = p_2'$

$\vartheta_3 = 780°C = 1053,15K$

$\eta_{ik} = 0,83$

$\eta_{it} = 0,85$

$$\frac{T_1}{T_2} = \left(\frac{p_1}{p_2}\right)^{\frac{k-1}{k}} \Rightarrow T_2 = 456,38K$$

$$\boxed{c_p = c_v \cdot K = 1004,5\ J/kgK}$$

$c_v \cdot K = c_v + R$

$c_v(K-1) = R$

$c_v = \frac{R}{K-1} = 717,5$

$$\frac{T_3}{T_4} = \left(\frac{p_3}{p_4}\right)^{\frac{k-1}{k}}$$

$T_4 = 604,94K$

b) $W_{turb} = ?$

Turbina:

$W_{id} = W_{34} = h_3 - h_4 = c_p(T_3 - T_4)$

$W_{red} = W_{34} = h_3 - h_4 = c_p(T_3 - T_4)$

$$\boxed{W_t = 389,96\ kJ/kg}$$

c) $W_{kompr} = W_{12} = (h_1 - h_2) = c_p(T_1 - T_2) = -168,99\ kJ/kg$

d) $q_{dov} = q_{23} = h_3 - h_2 = c_p(T_3 - T_2) = 599,46\ kJ/kg$

e) $\eta = ?$

$$\eta = \frac{W_k + W_t}{q_{dov}} = 0,369$$

(13) PWR

$n = 4$

$m_{UO_2} = 105\,t$

$e = 0{,}03$

$\phi = 3 \cdot 10^{17}\ n/m^2 s$

$\sigma_f = 580 \cdot 10^{-28}\ m^2$

chładziwo:

$c_p = 5{,}7\ kJ/kgK$

$\rho_{ch} = 720\ kg/m^3$

$T_{wl} = 295°C$

$T_{izl} = 328°C$

$P_T = 3980\ MW$

$h_{FW} = 391\ kJ/kg$ (pojna woda)

$h_{ST} = 2764\ kJ/kg$ (zasicona para)

$$\boxed{P_T = \dot{m}_{pare} \cdot \Delta h = P_j + n \cdot P_p}$$

$$P_p = \dot{m}_{chlad} \cdot W_p = \dot{m}_{chlad} \cdot V \cdot \Delta p_p$$

$$P_j = \dot{m}_{chlad} \cdot c_v \cdot \Delta T \cdot n$$

a) $P_j = ?$

b) $\dot{m}_{chladiwo} = ?$

c) $\dot{m}_{pare} = ?$

d) $P_p = ?$

e) $\Delta P_p = ?$

a) $P_j = 3{,}962\ GW$

$N_{U235} = 7{,}115 \cdot 10^{27}$

b) $\dot{m}_{chlad} \Rightarrow$

~~$P_p = \dot{m}_{chlad} \cdot V \Delta p_p$~~

$P_j = \dot{m}_{chlad} \cdot c_v \cdot \Delta T \cdot n$

$\dot{m}_{chlad} = 31252{,}64\ kg/s$

c)

$P_T = \dot{m}_{pare} \cdot \Delta h$

$\dot{m}_{pare} = 1677{,}2\ kg/s$

d) $P_p = \frac{-P_j + P_T}{4}$

$P_p = 4{,}5\ MW$

e)

$$P_p = \dot{m}_{chlad} \cdot \frac{1}{\rho} \cdot \Delta p_p$$

$\Delta p_p = 103{,}67\ kPa$

(14) Tlačni tunel:

$H_{zah} = 100\,m$ n.v.

$H_t = 40\,m$ n.v.

$H_{GV} = 120\,m$ n.v.

$H_{DV} = 32\,m$ n.v.

η -> nije zadano => 1

a) P_T, $Q = 100\,m^3/s$

- bez difuzora

$D = 3m$

b) P_T, $Q = 100\,m^3/s$,

- DIFUZOR

$D = 5m$

BEZ DIFUZORA

a) $\boxed{H_w = H_{GV} - H_t - \frac{c^2}{2g}} = \underline{69{,}799\,m}$

$Q = c \cdot A$

$c = \frac{Q}{A} = 14{,}147\,m/s$

$A = \frac{D^2 \pi}{4} = 7{,}069\,m^2$

$\boxed{P = 68{,}47\,MW}$

b) SA DIFUZOROM

$A = 19{,}635\,m^2$

$c = 5{,}093\,m/s$

$\boxed{H_{wa} = H_{GV} - H_{DV} - \frac{c^2}{2g}} = \underline{86{,}678\,m}$

$\boxed{P = 85{,}03\,MW}$

(15)

(5) Solarni toranj i FN ćelije

$G = 1200\ W/m^2$

$H_e = 1500\ kWh/m^2$

$A_o = 10000\ m^2$

$\eta_S = 0{,}5$

$\eta_R = 0{,}3$

$\eta_{FN} = 0{,}11$

$\eta_{DIR} = 0{,}8$

a) $P_i = ?$

b) $W_{el,god} = ?$

$\frac{H_{Bopt}}{H_e} = 1{,}20$

$\eta_{usmj} = 1{,}2$

c) $m = ?$

FN:

$W = H_e \cdot A \cdot \eta \cdot \frac{H_{Bopt}}{H_e}$

Sol. toranj	FN-ćel.
a) $P = P_{direct} \cdot \eta_S \cdot \eta_R$ $= P_n \cdot \eta_{DIR} \cdot \eta_S \cdot \eta_R$ $= A \cdot G \cdot \eta_{DIR} \cdot \eta_S \cdot \eta_R$ $= 1{,}44\ MW$	$P = G \cdot A_{FN} \cdot \eta_{FN}$ $= 1{,}32\ MW$
b) $\eta_{usmj.} = 1{,}2$ $W = A \cdot H_n \cdot \eta_{usmj} \cdot \eta_S \cdot \eta_{el,pot}$ $= 2{,}16\ GWe$	$W = H_e \cdot A \cdot \eta \cdot \frac{H_{Bopt}}{H_e}$ $= 1{,}98\ GWe$
d) $m = 0{,}17123$	$= m = 0{,}17123$

$m = \frac{W}{T \cdot P}$ godina!

(17.) EES

t(h)	0-4	4-6	6-9	9-12	12-14	14-18	18-22
P(MW)	800	1000	1100	2000	1700	1500	1700

22-23	23-24
1200	800

a) $\boxed{W_{k} = 19200\,MWh}$

$P_V = 1200\,MW$

$\boxed{W = 32000 + \dots = 27600\,MWh}$

$\boxed{W_V = 8400\,MWh}$

$m = \frac{W}{24h \cdot P_{max}}$

$\boxed{m = 0{,}575}$

HE 1: 200MW } protočn.

HE 2: 300MW

NE: 500MW

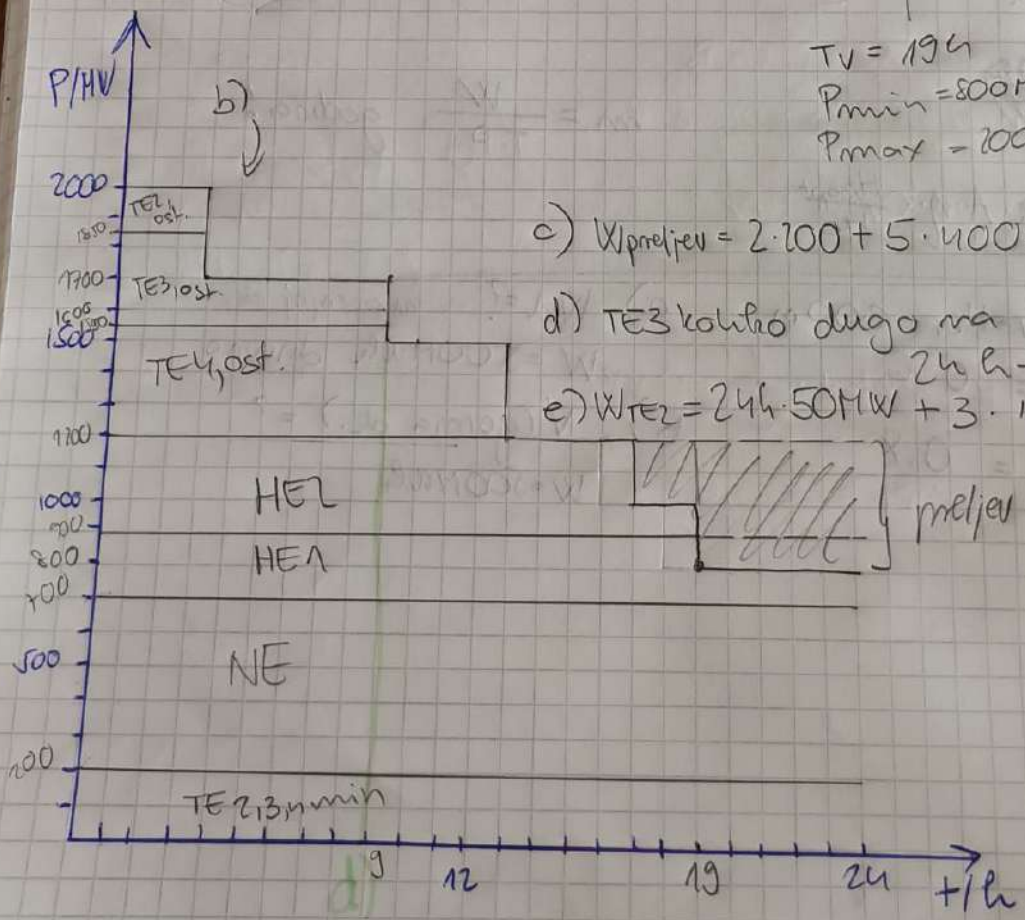
	ne palim	min	cij:
TE1:	200MW	50MW	40 €/...
TE2:	250 -"- (palim 150 min)	50MW	35 ...
TE3:	350 -"-	50MW	30 -"-
TE4:	450 -"-	100MW	25 -"-

$T_V = 19h$

$P_{min} = 800\,MW$

$P_{max} = 1000\,MW$

b)



c) $W_{prelijev} = 2 \cdot 200 + 5 \cdot 400 = \underbrace{2400\,MWh}$

d) TE3 koliko dugo na min snazi?

$24h - 9h = \boxed{15h}$

e) $W_{TE2} = 24h \cdot 50MW + 3 \cdot 150 = \underline{1650\,MWh}$